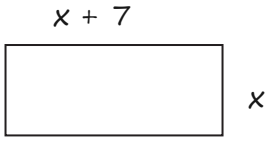
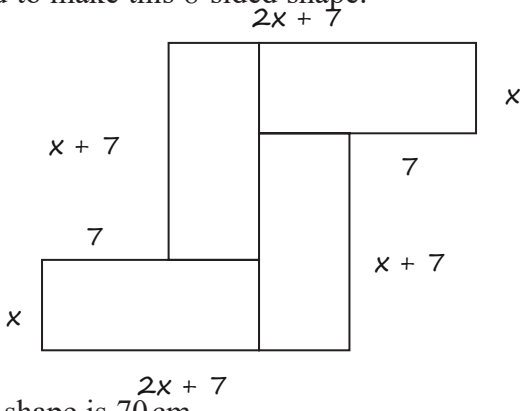
24 Here is a rectangle.



1 mark
Width x; length x + 7

The length of the rectangle is 7 cm longer than the width of the rectangle.

4 of these rectangles are used to make this 8-sided shape.



1 mark
Identify the outside sides of the complete shape

The perimeter of the 8-sided shape is 70 cm.

Work out the area of the 8-sided shape.

Perimeter = (2x + 7) + x + 7 + (x + 7)

+ (2x + 7) + x + 7 + (x + 7)

Collect the like terms: 8x + 42.

$8x + 42 = 70$

$-42 \quad -42$

$8x = 28$

$\div 8 \quad \div 8$

$x = 3.5$

1 mark
Equate the perimeter to 70

1 mark
Width 3.5 cm; length 10.5 cm

Each rectangle is 3.5 cm by (3.5 + 7) cm = 3.5 cm by 10.5 cm.

Total area = 4 × 3.5 × 10.5 = 147 cm^2

1 mark
Area of four equal rectangles

147 ..... cm^2

(Total for Question 24 is 5 marks)